

Data Management

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```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.4
```

```
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(descr)
library(knitr)
library(dplyr)
library(readr)
library(readxl)
```

```
#===My Data===
```

```
file_path <- "/cloud/project/data/data.xls"

data.xls <- read_excel(file_path)
options(width=80)
head(data.xls)
```

```
# A tibble: 6 x 138
  RespondentID Country A3.1 A3.2 A3.3 A3.4 A3.5 A3.6 A3.7 A3.8 A3.9
    <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1         1         1     2     3    52    52     4     1     5    12     1
2         2         1     2     5    52    52     3     1     8     2     5
```

```

3          3          1          1          4          94          94          4          1          5          2          1
4          4          1          1          2         221         221          1          1         10          3          4
5          5          1          1          3          52          52          3          1          8          2          0
6          6          1          1          1          52          52          1          4          9          3          4
# i 127 more variables: A3.10 <dbl>, A5.1 <dbl>, A5.2 <dbl>, `A5.3*` <dbl>,
#   A5.4 <dbl>, A5.5 <dbl>, A6.1 <dbl>, A7.1 <dbl>, A7.2 <dbl>, A9.1 <dbl>,
#   A9.2 <dbl>, `A9.3*` <dbl>, A9.4 <dbl>, A9.5 <dbl>, A9.6 <dbl>, A9.7 <dbl>,
#   `A9.8*` <dbl>, A11.1 <dbl>, A11.2 <dbl>, A11.3 <dbl>, A11.4 <dbl>,
#   A12.1 <dbl>, A12.2 <dbl>, A12.3 <dbl>, A12.4 <dbl>, A13.1 <dbl>,
#   A13.2 <dbl>, A13.3 <dbl>, A13.4 <dbl>, A14.1 <dbl>, A14.2 <dbl>,
#   A14.3 <dbl>, A14.4 <dbl>, A16.1 <dbl>, A16.2 <dbl>, `A16.3*` <dbl>, ...

```

===Frequency of Different Religious Orientations===

```
freq(as.ordered(data.xls$A3.8), plot = FALSE)
```

```

as.ordered(data.xls$A3.8)
      Frequency  Percent Cum Percent
1           293   14.4264    14.43
2           566   27.8680    42.29
3            87    4.2836    46.58
4            22    1.0832    47.66
5            11    0.5416    48.20
6            10    0.4924    48.70
7            11    0.5416    49.24
8            15    0.7386    49.98
9            55    2.7080    52.68
10           155    7.6317    60.32
11            75    3.6928    64.01
12           731   35.9921   100.00
Total        2031  100.0000

```

This frequency table shows the many different religious orientations that a sample included. What this is showing me is how many people are a part of a Catholic, Protestant or any other religious orientations. This will be important as I need to gather a variable in which excludes those who are not religious.

===Frequency of Levels of Educations===

```
freq(as.ordered(data.xls$A3.10), plot = FALSE)
```

```

as.ordered(data.xls$A3.10)
      Frequency  Percent Cum Percent
1            49    2.413    2.413
2           323   15.903   18.316
3           330   16.248   34.564

```

4	565	27.819	62.383
5	441	21.713	84.097
6	252	12.408	96.504
7	37	1.822	98.326
8	34	1.674	100.000
Total	2031	100.000	

This frequency table shows the levels of education within a sample. It shows me how many people don't have a high school degree to how many people have a doctorates. This will help me relate the info to people who are religious, to their levels of education. I need this info because I will eventually compare the information I gather to how many took COVID-19 as a serious threat.

===Frequency of who believed COVID-19 was exaggerated===

```
freq(as.ordered(data.xls$`A5.3*`), plot = FALSE)
```

```
as.ordered(data.xls$`A5.3*`)
```

	Frequency	Percent	Cum Percent
1	592	29.148	29.15
2	401	19.744	48.89
3	259	12.752	61.64
4	168	8.272	69.92
5	148	7.287	77.20
6	139	6.844	84.05
7	122	6.007	90.05
8	94	4.628	94.68
9	60	2.954	97.64
10	48	2.363	100.00
Total	2031	100.000	

This frequency table shows the levels in which how many people thought COVID-19 was being exaggerated. It goes from level 1 = Not at all to level 10= Extremely. This will help me see how seriously people took COVID-19 and who underestimated it.

===Frequency in which mask was worn===

```
freq(as.ordered(data.xls$A31.4), plot = FALSE)
```

```
as.ordered(data.xls$A31.4)
```

	Frequency	Percent	Cum Percent
1	874	43.033	43.03
2	121	5.958	48.99
3	120	5.908	54.90
4	185	9.109	64.01
5	184	9.060	73.07
6	195	9.601	82.67
7	352	17.331	100.00
Total	2031	100.000	

This frequency table shows how many people had frequently put on a mask during COVID-19. It goes from level 1, Never, to level 7, Always. This will help aid my question on who was taking COVID-19 seriously.

===Distance Away Frequency===

```
freq(as.ordered(data.xls$A31.6), plot = FALSE)
```

```
as.ordered(data.xls$A31.6)
      Frequency  Percent Cum Percent
1             24   1.1817     1.182
2             20   0.9847     2.166
3             36   1.7725     3.939
4            106   5.2191     9.158
5            138   6.7947    15.953
6            496  24.4215    40.374
7           1211  59.6258   100.000
Total         2031 100.0000
```

This table shows the frequency in which a person kept a distance of 1.5 meters away from others. The chart goes from level 1, Never, to level 7, Always. This will help me see who didn't wear their masks during COVID-19, which relates to my question of who didn't follow the guidelines provided to everyone across the globe.

=== Variable for Levels of Education===

```
data.xls$LVL_EDU <- factor(ifelse(data.xls$A3.10 %in% c("1","2"), "High School and Below", "Co
summary(data.xls$LVL_EDU)
```

```
College and Above High School and Below
      1659      372
```

I am creating a variable named LVL_EDU to help me see who have educations from up to High School then College and higher.

===Variable for People of Only Religious Orientations===

```
data.xls$ONLY_R <- factor(ifelse(data.xls$A3.8 %in% c("1", "2","3","4","5","6","7","8","9","11
summary(data.xls$ONLY_R)
```

```
Has a Religious Belief      No Beliefs
      1145      886
```

I am creating a variable named ONLY_R to group up who and who don't have a religious orientation.

===Variable for People Who Think COVID-19 was over exaggerated===

```
data.xls$CVD_EXAG <- factor(ifelse(data.xls$`A5.3*` %in% c("1", "2","3","4","5"), "Reasonable Concern", "Out-of-Proportion Reasonable Concern"))
summary(data.xls$CVD_EXAG)
```

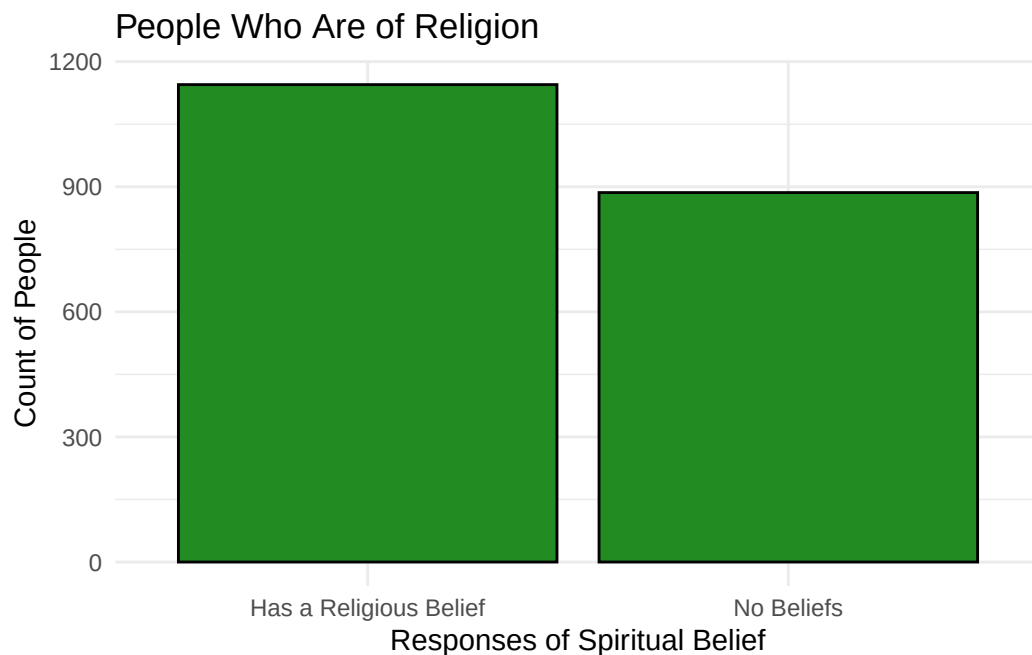
```
Out-of-Proportion Reasonable Concern
463                                1568
```

I am creating the variable CVD_EXAG to summarize the responses 1 = Not at all, 10= Extremely into two managed variables, options 1-5 being considered the people who believed that COVID-19 was not being over-exaggerated and options 6-10 being considered the people who thought the virus was being told to be a larger issue than it really was.

===Univariate Graphs===

```
library(ggplot2)

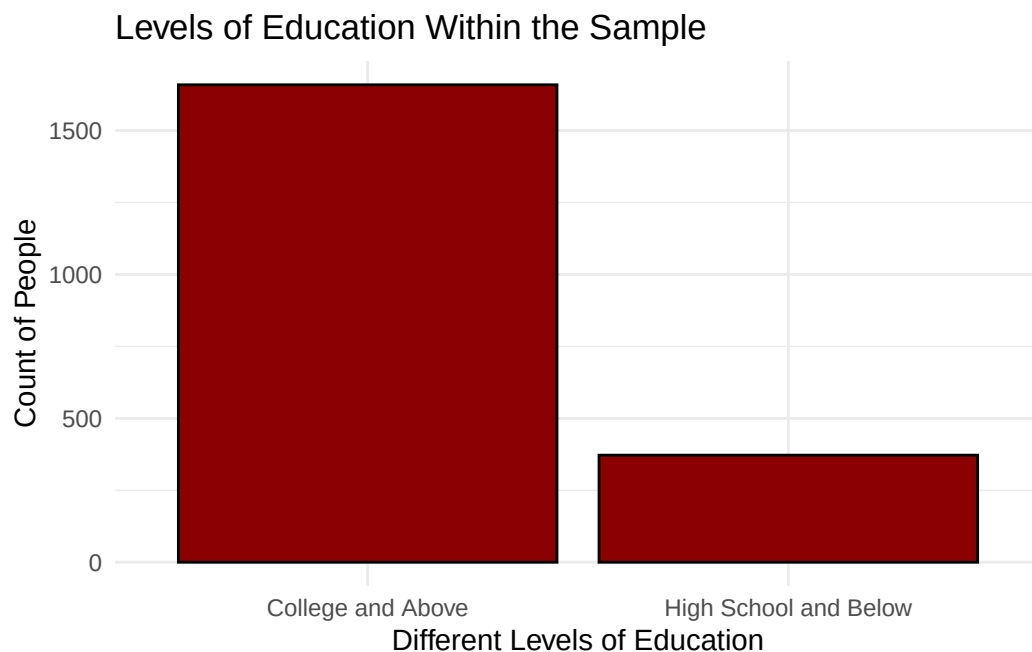
data.xls %>%
  ggplot(aes(x = ONLY_R)) +
  geom_bar(fill = "forestgreen", color = "black") +
  ggtitle("People Who Are of Religion") +
  xlab("Responses of Spiritual Belief") +
  ylab("Count of People") +
  theme_minimal()
```



I am developing this graph to show how many people in the sample do and do not have any religious perspectives. I would like to correlate this graph later on with my other variables to see if there are any patterns between each other. One of my main concerns is that some people of religious backgrounds didn't think COVID-19 was a problem and that nothing would happen to each other. With this data gathered and graphed, I think it'll become easier to start comparing.

```
library(ggplot2)

data.xls %>%
  ggplot(aes(x = LVL_EDU)) +
  geom_bar(fill = "darkred", color = "black") +
  ggtitle("Levels of Education Within the Sample") +
  xlab("Different Levels of Education") +
  ylab("Count of People") +
  theme_minimal()
```

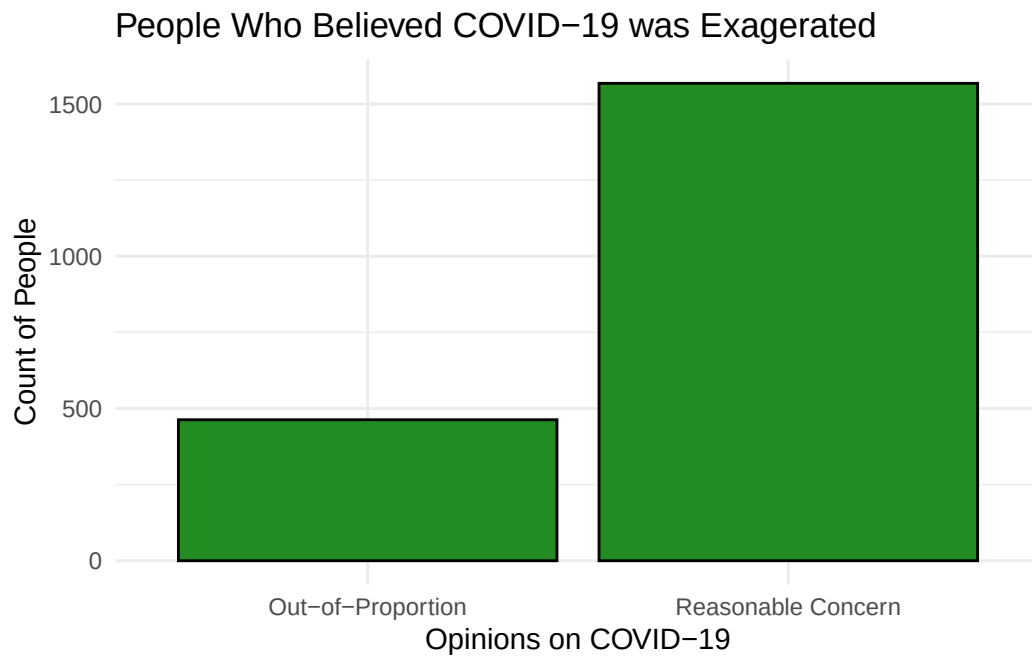


This graph will help to visualize the differences between two major levels of education. The goal of this is to understand whether or not the levels education dictate what people believed during the COVID-19 pandemic and how serious the virus was. My graph here is just to show how many in the sample have received higher education or not.

```
library(ggplot2)

data.xls %>%
  ggplot(aes(x = CVD_EXAG)) +
  geom_bar(fill = "forestgreen", color = "black") +
  ggtitle("People Who Believed COVID-19 was Exaggerated") +
  xlab("Opinions on COVID-19") +
```

```
ylab("Count of People") +  
theme_minimal()
```



This graph shows the many people who believed whether COVID-19 was overtly told as a massive issue and the many others who believed the concern of the virus was authentic. The main goal for this chart is to get an idea of the opinions many had about the virus when the issue was most prevalent.